

COLLARDRIVE · PRECIS

CollarDrive design precis

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
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Source	docs/02-concept.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-03	Amish Chadha	TRL 3 numbers from CLD-CAL-001 v0.2 on the constructable design (CLD-DDR-002); key design choices decided under Amish's 2026-10-03 pre-approval (CLD-DDR-001)
0.2	2026-10-03	Amish Chadha	TRL 2 update with concept media, first-order numbers and key design choices
0.1	2026-09-30	Amish Chadha	Initial scaffold

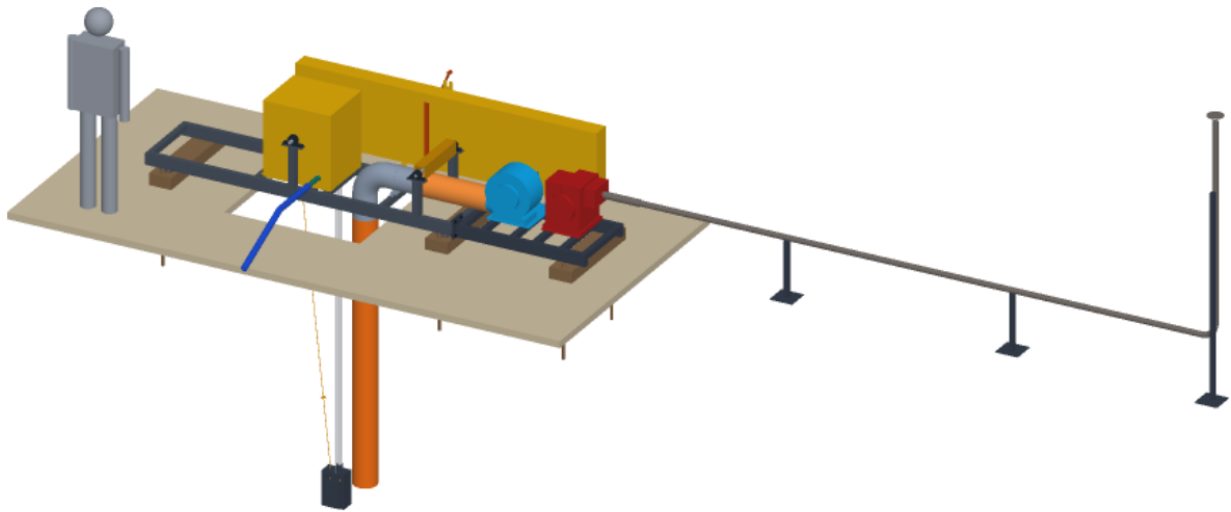
CollarDrive design precis

Keeps engines on the surface by driving a rope pump and blower from a frame on the shaft collar.

CollarDrive

github.com/BoujeeEnjinia1701/collardrive

CONCEPT, NOT FOR FABRICATION



Seen from the front right and above, 26 deg elevation. Grey figure: 1.75 m person for scale, standing beside the collar.
Pipe, rope and duct drawn shortened; exhaust riser at right

Figure 1. CollarDrive on its sleepers over a shaft collar, with a 1.75 m person for scale. Pipe, rope and duct drawn shortened.

■ Summary

CollarDrive is a two-piece steel frame that sits on timber sleepers over an artisanal mine shaft. One small petrol engine on the frame drives a layshaft, which turns a rope pump wheel and a ventilation blower through guarded V-belts. Only the rope, the rising main and an air duct go down the shaft; the exhaust leaves through a pipe and a riser more than 6 m downwind. On paper it lifts about 76 L/min from 20 m and 19 L/min from 60 m and delivers 0.19 m³/s of fresh air at the end of 30 m of duct, using under 0.8 kW of the engine's power. The estimated kit cost is USD 2,535 against a USD 3,500 value-engineering target.

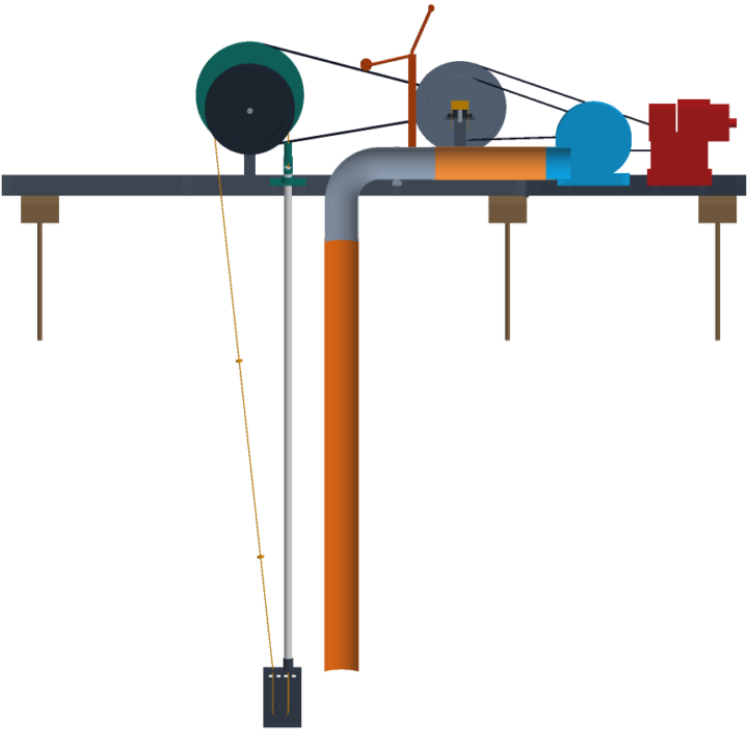
■ How it works

The engine sits on a slotted plate on the drive module, beyond the shaft opening. Its 80 mm pulley drives a 480 mm pulley on a layshaft at 467 rpm. From the layshaft, two belts turn a 560 mm pulley on the rope pump shaft at 75 rpm, and one belt turns the blower at about 2,000 rpm. The pump wheel's rubber V grips a 6 mm rope carrying a plastic piston every metre. The rope runs down the open side of the shaft to a weighted guide block at the bottom, turns round a roller, and rises through the PVC main, pushing a column of water ahead of the pistons. The water leaves by a tee and a spout at the frame and runs away down a hose. The blower pushes fresh air through a rigid bend and down a 200 mm lay-flat duct to the working level. A lever on the belt guard slackens the pump belts, so air can run without pumping. A ratchet on the pump shaft stops the water column running the rope back when the engine stops. The exhaust goes through a flexible hose and a pipe on stands to a riser 2.4 m high, 7 m from the collar edge.

CollarDrive: cutaway

github.com/BoujeeEnjinia1701/collardrive

CONCEPT, NOT FOR FABRICATION



Cut on the frame centre line, front half removed, guards and exhaust left out; seen from the front, 12 deg elevation.
Rope up the rising main, over the wheel, down to the guide block; duct beside it

Figure 2. Cutaway on the frame centre line: rope up the main, over the wheel and down to the guide block, with the duct beside it.

■ Main components

Table 1. Components (numbers match bom/bom.csv).

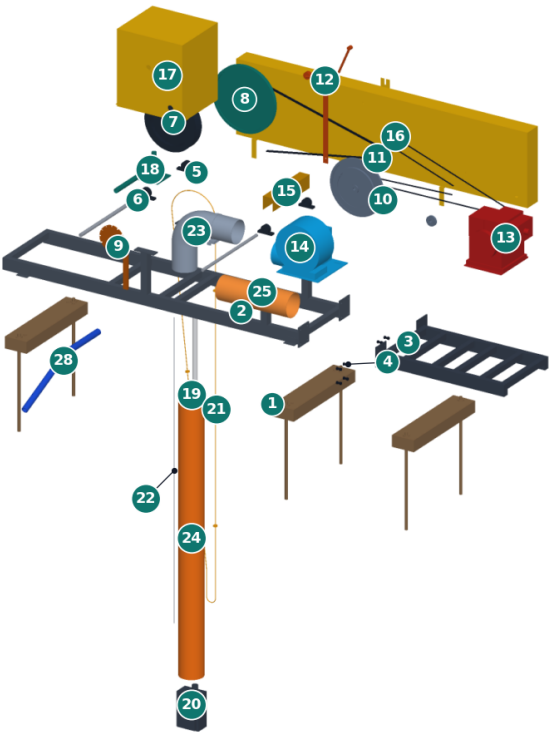
BOM	Component	Role
1	Sleepers, stakes and coach screws	Carry the frame on the ground, clear of the collar lip
2	Span module with pedestals and cleats	Bridges the opening; carries the wheel, the layshaft, the pipe head and the duct head
3, 4	Drive module and joint bolts	Carries the engine and the blower; bolted to the span module
5, 6	Pillow blocks and shafts	30 mm pump shaft and 25 mm layshaft
7, 8, 9	Rope pump wheel, pump pulley, backstop	Drive the rope; stop it running back
10, 11, 12	Pulleys, V-belts, clutch idler	Engine to layshaft, layshaft to pump and to blower; pump on and off
13, 14	Engine and blower on slotted plates	Power and fresh air
15, 16, 17	Layshaft cover, belt guard, wheel hood	Keep hands and clothing out of moving parts
18 to 22	Pipe head, rising main, guide block, rope and pistons, support wire	The rope pump in the shaft
23 to 25	Duct head, lay-flat duct, flexible link	Air from the blower down the shaft
26, 27	Exhaust extension and stands	Exhaust 6 m or more from the collar and the intake
28, 29	Discharge hose, gas detector	Water away from the collar; gas check below ground

CollarDrive: exploded view

github.com/BoujeeEnjinia1701/collardrive

CONCEPT, NOT FOR FABRICATION

- 1 Sleepers, stakes and coach screws
- 2 Span module with pedestals and cleats
- 3 Drive module
- 4 Module joint bolts
- 5 Pillow block bearings
- 6 Pump wheel shaft and layshaft
- 7 Rope pump wheel
- 8 Pump pulley
- 9 Backstop ratchet and pawl
- 10 Pulleys
- 11 V-belts
- 12 Clutch idler and lever
- 13 Engine and engine plate
- 14 Blower, plate and inlet guard
- 15 Layshaft cover
- 16 Belt guard
- 17 Pump wheel hood
- 18 Pipe head, tee and spout
- 19 Rising main (drawn shortened)
- 20 Guide block
- 21 Rope and pistons
- 22 Support wire
- 23 Duct head and saddle
- 24 Lay-flat duct (drawn shortened)
- 25 Flexible duct link
- 28 Discharge hose



Seen from the front right and above, 24 deg elevation; numbers match bom/bom.csv. Exhaust extension (26, 27) not shown

Figure 3. Exploded view with BOM numbers (exhaust extension not shown).

■ Numbers from the TRL 3 calculations

Table 2. Key results (CLD-CAL-001 v0.2; estimates).

Quantity	Value	Tag
Engine speed, layshaft, rope wheel, blower	2,800, 467, 75 and 2,007 rpm	A1
Rope speed	1.57 m/s	A2
Water at 20 m (40 mm main)	75.7 L/min	B20
Water at 30 m (40 mm main)	69.8 L/min; pump belts at 92 % of allowable pull	B30
Water at 60 m (25 mm main)	19.3 L/min	B60
Air at the end of 30 m of duct	0.189 m ³ /s, about four people at 0.05 m ³ /s each	C30
Engine shaft power at 30 m	0.78 kW, 3.8 times margin on 3.0 kW continuous; 0.58 L/h of petrol	D30
Exhaust outlet from collar edge and intake	7.05 m and 6.46 m	E1
Frame stress, working and misuse	27 MPa and 47 MPa (S275)	F2, F3
Ground bearing	About 7 kPa	F4
Heaviest piece	Span module, 69.6 kg	H1
Estimated cost	Value-engineering target: USD 3,500. Estimated cost of the constructable design: USD 2,535 (USD 965 under the target)	H2

CollarDrive: power flow at 20 m head, pump and blower running (all values are estimates)

github.com/BoujeeEnjinial701/collardrive

CONCEPT, NOT FOR FABRICATION

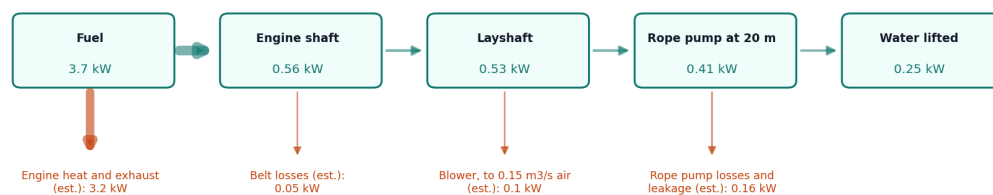


Figure 4. Power flow at 20 m head (estimates).

■ Key design choices

All decided by Amish under his pre-approval of 2026-10-03 (CLD-DDR-001 and CLD-DDR-002); see the design decisions register, CLD-DEC-001.

- **One surface engine, one layshaft.** A 4.8 kW class petrol engine governed to 2,800 rpm; a diesel of similar size fits the same plate.
- **Two belts on the pump stage.** One belt would be overloaded at 30 m.
- **Idler clutch.** Air can run alone; nothing to buy.
- **Rope pump sized by depth.** 40 mm main to 30 m, 32 mm to 45 m, 25 mm to 60 m.
- **Forcing ventilation** through 200 mm lay-flat duct, intake at the front of the frame.

- **Exhaust riser downwind**, at least 6 m from the collar and the intake.
- **Two carried modules on set-back sleepers**, nothing bearing on the collar lip.
- **Backstop, full guards and a gas detector** in every kit.

■ Patent design-arounds

From the preliminary patent, trademark and prior-art screen (not legal advice):

- The documentation cites rope pump heritage: the public-domain rope and washer pump and the Practica motorised rope pump.
- No patent risk was found in the preliminary screen; the engineering risks noted, pump output at depth and duct pressure loss, are covered in CLD-CAL-001 sections B and C.

■ Shared blocks

- Surface power take-off block: the engine plate, layshaft and idler clutch, offered as a shared block for other lab projects that need one engine to drive two machines.
- FieldNode sensor core: a possible carbon monoxide and oxygen monitor at the shaft bottom, later.
- CalRig proof-load: a frame proof test at TRL 4.

■ Safety

Safety: CollarDrive is published as an open engineering reference, not as certified mine ventilation or pumping equipment.

- Never run any engine in the shaft. Keep the exhaust riser downwind and at least 6 m from the collar and the blower intake (NIOSH advises at least 6 m (20 ft) from openings).
- Ventilation does not make a shaft safe on its own. Use the gas detector (carbon monoxide and oxygen) before and during work below ground.
- Moving machinery: belts, pulleys, shafts, the wheel and the rope can catch hands, hair and clothing. Run only with every guard fitted; stop the engine before touching the drive or the rope.
- The ratchet backstop must be fitted: without it the water column can run the rope and wheel backwards when the engine stops.
- The frame covers an open shaft. Fence the collar and never stand or kneel on the frame over the opening.
- Hot exhaust parts: the hose and pipe near the engine burn; keep them guarded by distance and do not touch them until cool.
- Refuel only with the engine stopped and cool, away from the collar.

This design is published as an open engineering reference. It is not certified equipment.

■ Open questions

None for TRL 3. Every decision is recorded in the design decisions register (CLD-DEC-001); facts that can only be settled with real parts are listed there under "To confirm when parts are bought".